Motif Search Problem

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gatc

acaagatcgacttgcaccg

cacgtacgaggtgactaac

cgattcaggactaggta

gatc

acaagatcgacttgcaccg

cacgtacgaggtgatcaac

cgattcaggatcaggta

gatc

acaagctcgacttgcaccg

cacgtacgaggtaatcaac

cgattcaggattaggta

acaagctcgacttgcaccg
cacgtacgaggtaatcaac
cgattcaggattaggta

Planted (ℓ, d)-Motif Problem

Problem Definition

Given: Set of strings $S=\{s_1, s_2, \ldots. S_n\}$, $\sum = \{A, C, G, T\}$, *ℓ and d*.

Objective: To find all strings *x* such that
$|x| = \ell$ and *x* occurs in s_i, for $1 \leq i \leq n$
with a Hamming distance of *d*.

- Step 1: For each string i, find all substrings of length L and store in an array S_i. [Total number of substrings = n*(m-L+1)
- Step 2: For each array S_i , Generate neighborhood (efficiently) of each substring in the array. Let's call it T_i. [Total number of neighbors = $\binom{L}{D}3^D$]
- Step 3: Find all substrings x that occur in each of the T_i's.
    - Option 1 – Sort each of the T_i's and look of repeated patterns.
    - Option 2 – Hashing
    - Option 3 – Sets
    - Option 4 – Anything you propose
  - For testing , show me the screen shot for L = 15, D=5, N = 20, M = 600 (length of each sequence)

acg

Neighborhood of acg: D = 1

Location 1: gcg, tgc, ccg

Location 2: agg, atg, acg

Location 3: acc, act, aca

Choose D locations from L, at each location you have 3 options.

$$\binom{L}{D}3^{D}$$